

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fifth Semester of B. Tech (CE) Examination****May 2017****CE313 System Software****Date: 24.05.2017, Wednesday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections and both must be attempted in separate answer sheets.
2. Make suitable assumptions and draw neat figures wherever required.
3. Write entire option in case of multiple choice and matching questions.

SECTION – I**Q – 1 Answer the following short questions.**

- (A) Define following terms. [04]
 (a) System Software (b) System Programming (c) Process (d) Job
- (B) List phases of compiler. [03]
- (C) Explain DC and DS statements with example. [02]
- (D) What is backpatching? Why it is needed? [02]

Q - 2 Answer the following questions. (Any three)**[12]**

- (A) Explain preprocessor and macro processor.
- (B) Briefly explain absolute loader and dynamic loader.
- (C) What are the methods available for debugging in Turbo C++?
- (D) List the data structures used in macro processing.

Q – 3 Answer the following questions.

- (A) What is editor? Write different types of editor. [03]
- (B) Differentiate compiler and interpreter. [04]
- (C) Explain data structures used in two pass assembler. [05]

OR**Q – 3 Answer the following questions.**

- (A) What do you mean by program relocation? What so you mean by self-relocatable program? [03]
- (B) What is nested macro? Explain how the nested macro call is processed? How stack is used to manage the data structures of macro in nested call? [04]
- (C) Explain Java interpreter. [05]

SECTION – II**Q - 4 Answer the following short questions.**

- (A) Write a regular expression for string over 0 and 1 starting with 0 and ending with 1. [01]
- (B) List out the tools to implement Lexical Analyser and Syntax Analyser. [02]
- (C) Write two difference between top-down and bottom-up parsing methods. [02]

- (D) What do you mean by ambiguous grammar? Show that following grammar is ambiguous. [03]
 $E \rightarrow E + E \mid E * E \mid a$
- (E) Why are we using LTORG statement? What happens if we have not used LTORG statement in our program? [03]

Q - 5 Answer the following questions. (Any three) [12]

- (A) Explain following assembler directives
(a) START (b) EQU (c) ORIGIN (d) END
- (B) List and explain strategies for resource allocation.
- (C) What is assembly language? List and explain different type of assembly statements.
- (D) What is symbol table? What operations can be performed on symbol table? Which data structures can be used to implement the symbol table? Which one of them is best data structure to implement the symbol table?

Q - 6 Answer the following questions.

- (A) Draw process state diagram. [03]
- (B) What do you mean by left factoring? Give example of left factored grammar. Also explain how to remove it from the grammar with example. [04]
- (C) Write a short note on Batch Processing Operating System. [05]

OR

Q - 6 Answer the following questions.

- (A) What is Language Processor? List the different types of language processors. [03]
- (B) Define Grammar. [04]
- (C) Draw a DFA for $(0|1)^*11$ [05]
